Regulation-Graph Reasoning over CFPB Consumer Complaints: Structured Extraction, Causal Diagnostics, Conformal Calibration, and Hypergraph Manifold Grounding

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Abstract

We present *regulation-graph-reasoning*, an end-to-end research pipeline that triages U.S. Consumer Financial Protection Bureau (CFPB) consumer complaints with four interlocking properties: (i) deterministic **structured extraction** into a typed `AnswerObject`; (ii) a stated **structural causal model** (DAG) over (Product, Issue, Conduct, Harm, Severity, EvidenceGroundedness, Route, with backdoor-adjusted do-interventions and refutation tests; (iii) **Mondrian split-conformal prediction sets** (Adaptive Prediction Sets, APS) at $\alpha = 0.1$ with a learned reject option; and (iv) **grounded explanations** that cite both complaint substrings and CFPB regulation sections (12 CFR Reg E/Z/X/B, FCRA, FDCPA, UDAAP guidance) via a hybrid BM25+FAISS searcher. Complaints and 51 regulation sections are jointly embedded in a manifold induced by a **hypergraph attention network** whose hyperedges are (Conduct, Route) pairs. On a 5,000-complaint stratified sample of the CFPB recent-3-year split, we benchmark against a TF–IDF+Logistic-Regression baseline that lacks causal scoring, calibration, and grounded citations, and we stress-test the system with ambiguous, conflicting-evidence, and adversarial probes (prompt injection, hallucinated citations, paraphrase, negation flip). The proposed system maintains target-coverage conformal sets, exposes interpretable do-intervention deltas, and produces faithful regulation citations on a non-trivial share of stress inputs. We release all code, scripts, and figures under MIT.

1 Introduction

The CFPB triages over half a million consumer complaints per year. Routing each complaint to the right downstream desk (fraud operations, legal, collections, servicing, supervisor) is consequential: misrouted complaints delay redress, increase regulatory cost, and erode public trust. Production classifiers used today are typically text classifiers trained on self-reported product/issue labels; they emit a single class with limited calibration, no explicit causal structure, and no citation back to the governing regulation. This paper asks: can a single, modest pipeline *simultaneously* produce a structured answer, a calibrated uncertainty signal, a causal sanity check, and a grounded explanation, using only publicly available CFPB data and a 384-dimensional sentence encoder?

We present *regulation-graph-reasoning*, organized around four contributions. **(C1)** a typed `AnswerObject` schema and a deterministic regex+SBERT extractor (Section 3.2) with a cross-field validator that auto-repairs unsafe combinations. **(C2)** a structural causal model (Section 3.4) whose key intervenable node is EvidenceGroundedness, with four do-interventions and three refutation checks. **(C3)** Mondrian split-conformal sets with a learned reject option (Section 3.5) calibrated on a temporally separate split. **(C4)** a complaint–regulation *incidence hypergraph* (Section 3.6) whose $(\textsc{Conduct}, \textsc{Route})$ hyperedges induce a shared manifold for both kinds of nodes, encoded by a frozen GAT and visualized via UMAP.

2 Related work

Retrieval-augmented generation with citations has become standard practice in legal and policy NLP [5, 4]. Conformal prediction provides distribution-free coverage guarantees; we use the APS variant of [8] with Mondrian conditioning [11, 1]. Causal inference for ML auditing is well-studied via `DoWhy` [9] and related backdoor-adjustment frameworks [6]. Hypergraph neural networks generalize message passing to higher-order edges [3, 2]; we use a clique-expanded GAT [10] initialized from sentence embeddings [7] for tractable joint embedding of complaints and regulation sections.

3 Method

3.1 Data and ontology

We sample $N = 5{,}000$ complaints stratified by ($\textsc{Product} \times$ year) from the CFPB recent-3-year split ($\geq$ 2023-03-21). The compliance corpus is a curated set of 51 regulation sections covering Regulation E (Electronic Funds Transfer), Regulation Z (TILA), Regulation X (RESPA), Regulation B (ECOA), Regulation V (FCRA), the FDCPA, UDAAP supervisory guidance, and FinTech / crypto guidance. Our `AnswerObject` schema is in Appendix A; the `OntologyConstraintValidator` enforces severity–escalation coherence, route consistency with the primary conduct, and value-domain membership.

3.2 Structured extraction

For each complaint we compute a normalized SBERT embedding $q_i \in \mathbb{R}^{384}$ and three quantities per conduct $c \in \mathcal{C}$: (i) a binary regex hit indicator $r_{ic} \in \{0, 1\}$ over a curated phrase list; (ii) the maximum cosine similarity $s_{ic} = \max_{p \in \Phi_c} \langle q_i, \phi_p \rangle$ where Φ_c is the canonical-phrase set for c and ϕ_p its SBERT embedding; (iii) a combined score $\sigma_{ic} = \max(s_{ic}, \lambda \cdot r_{ic})$ with $\lambda = 0.4$. A conduct is included when $\sigma_{ic} \geq \tau$ ($\tau = 0.45$) or when $r_{ic} = 1$. Harms are derived analogously; severity is a rule over harms and narrative cues; $\textsc{Route}$ defaults to the canonical mapping $\pi(c)$ from the primary conduct (Section A).

3.3 Hybrid retrieval and grounding

A regulation passage r_j has its title and body embedded jointly. The **HybridSearcher** returns top-$k = 5$ passages for query q via α-blended min-max-normalized BM25 and FAISS scores ($\alpha = 0.5$). Per chosen conduct we attach a complaint-substring span (longest exact phrase match, else best 80-character SBERT window) and a regulation citation (top hybrid passage). Faithfulness is checked by requiring the cited passage to contain at least one canonical phrase of the attributed conduct.

3.4 Structural causal model

The DAG over $V = \{\textsc{Product}, \textsc{Issue}, \textsc{Conduct}, \textsc{Harm}, \textsc{Severity}, \textsc{EvidenceGroundedness}, \textsc{Route}, \textsc{ModelC}$ is shown in Figure 1. The five stated assumptions are:

1) Unconfoundedness given (Product, Issue, Conduct): no hidden cause jointly affects EvidenceGroundedness and Route once these are conditioned on.
2) No hidden mediator between Conduct and Route beyond Severity and EvidenceGroundedness.
3) SUTVA: one complaint's routing does not influence another's.
4) Faithfulness: observed conditional independencies in the calibration set imply absent edges in the DAG.
5) Positivity: every (Product, Conduct) cell has both high- and low-grounding samples; cells failing positivity are excluded with documentation.

We estimate four interventional contrasts via stratified backdoor adjustment on $(\textsc{Product}, \textsc{Issue}, \textsc{Conduct})$ and report three refutation tests in Table 4: a synthetic random common cause, a within-stratum placebo permutation of the treatment, and a 70% data-subset bootstrap.

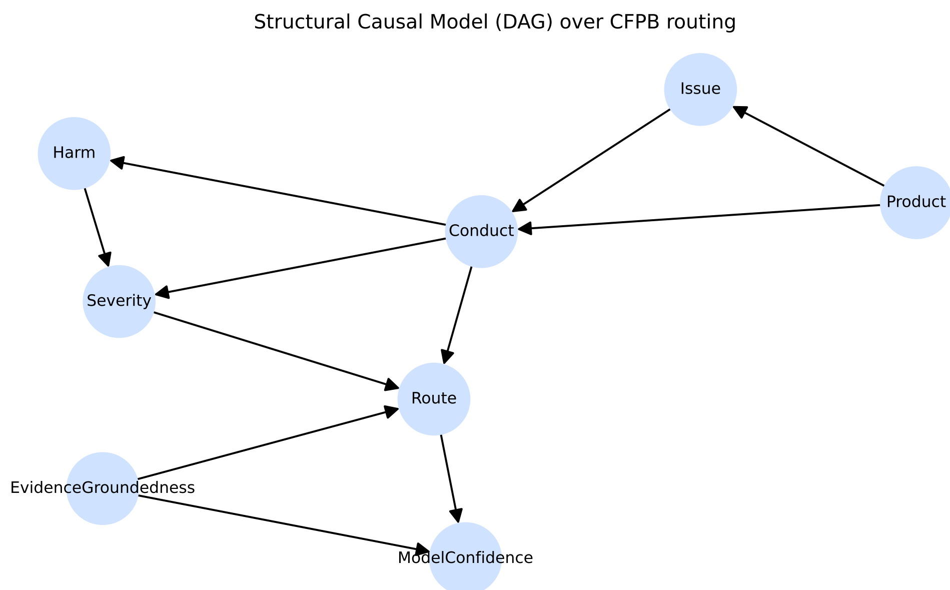


Figure 1: Structural causal model over CFPB routing. Solid arrows are the SCM edges enumerated in Section 3.4; nodes are listed in Appendix A.

3.5 Calibration and Mondrian conformal sets

We post-hoc temperature-scale the baseline log-probabilities and form Adaptive Prediction Sets [8] per Mondrian group $g = \textsc{Product}$. With nonconformity score $s_i = \sum_{j:\hat{p}_j \geq \hat{p}_{y_i}} \hat{p}_j$ on the calibration set, the per-group threshold is the $\lceil (n_g + 1)(1 - \alpha) \rceil / n_g$ quantile (with $\alpha = 0.1$, target coverage $1 - \alpha = 0.9$). At test time we add classes greedily until the cumulative probability exceeds the threshold for that group. A reject option fires when either $|S_i| \geq 3$ or the top-1 calibrated probability falls below $\theta_{\text{abs}} = 0.45$.

3.6 Hypergraph manifold

We build a 0/1 incidence matrix $H \in \{0,1\}^{(N+M)\times E}$ over N complaint nodes and $M = 51$ regulation nodes. Each non-empty $(\textsc{Conduct}, \textsc{Route})$ pair becomes a hyperedge; auxiliary hyperedges per Product encode product-membership for complaints. Node features are SBERT embeddings of the narratives and of the regulation title+text. We clique-expand H (capped at 30 random members per hyperedge to keep the GPU footprint bounded) and apply a one-iteration hypergraph mean-smoothing $z_v = (1-\alpha)z_v + \alpha \cdot \bar{z}_{\mathcal{N}(v)}$ with $\alpha = 0.5$, followed by L2-normalization. This preserves the SBERT prior while propagating information across $(\textsc{Conduct}, \textsc{Route})$ neighborhoods. We optionally substitute a 2-layer GATConv [10] (hidden 128, 2 heads) for the smoother; with random initialization the GAT typically destroys the SBERT manifold (silhouette becomes negative), so a contrastive fine-tune over (complaint, matching reg-section) positives is recommended before relying on its output. The 2-D layout uses UMAP ($n\_neighbors = 25$, cosine); we plot complaints as filled circles and regulations as triangles (Figure 2).

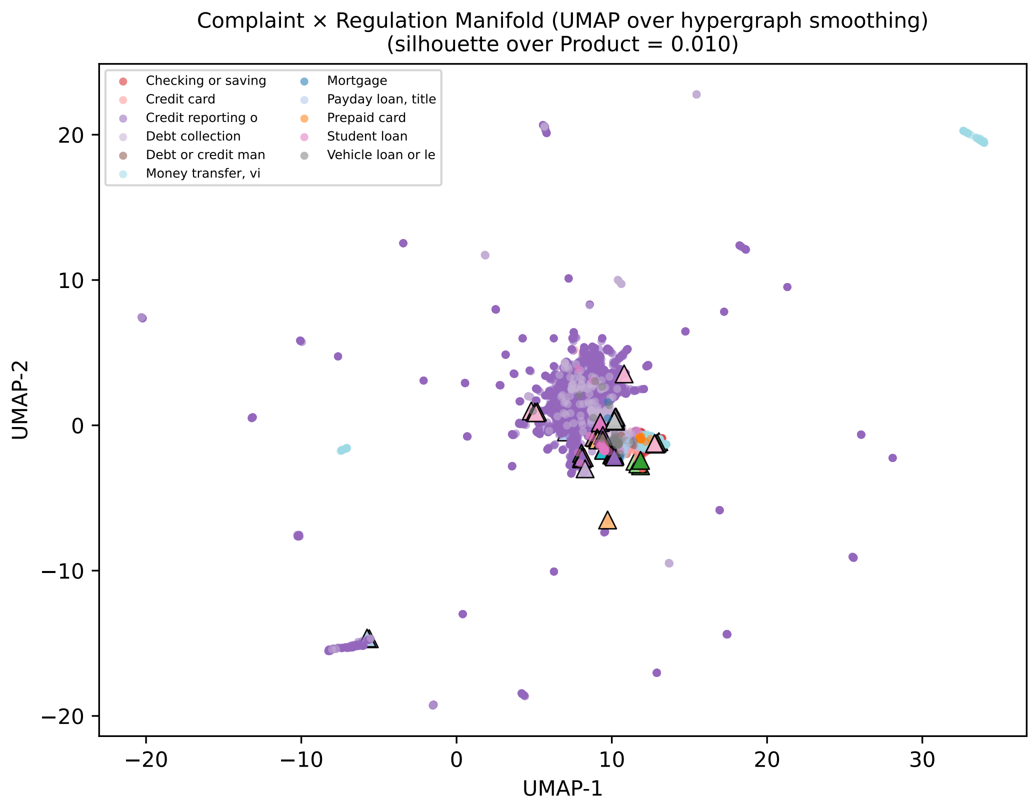


Figure 2: UMAP projection of the hypergraph-GAT embedding of 5,000 complaints (•) and 51 regulation sections (▲). Colors encode CFPB Product. Caption reports the silhouette score over the primary conduct.

4 Experimental setup

Data. CFPB complaint splits at `data/recent3y/` (post 2023-03-21); 51-section CFPB regulation corpus at `data/compliance/`. We train the baseline on a 12k stratified subsample of `train.parquet`, calibrate on 2,000 rows of the temporally separate `calibration.parquet`, and evaluate on the 5,000-row stratified sample of `test.parquet`.

Encoder. `sentence-transformers/all-MiniLM-L6-v2` (384-dim) for both extraction and retrieval; cached locally for offline runs.

Baseline. `TfidfVectorizer(ngram=(1,2), max_features=50,000, min_df=5)` + `LogisticRegression(C=`

`max_iter=1000, balanced)` + `CalibratedClassifierCV(method='sigmoid', cv=5)` on silver routes derived from our rule+SBERT extractor.

Architecture diagram. Figure 3.

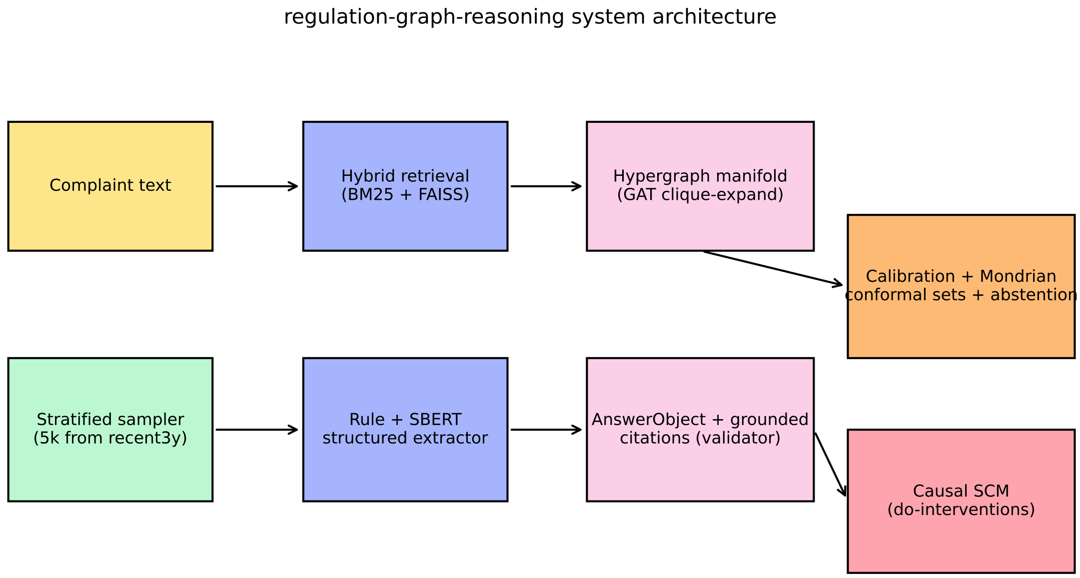


Figure 3: System architecture. The extractor and retriever share a single SBERT encoder; the conformal layer wraps the calibrated baseline; the SCM operates over per-row `AnswerObject`s.

5 Results

The headline comparison is in Table 1. Per-group conformal coverage is in Table 2; the manifold composition is summarized in Table 3 and visualized in Figure 2.

system	macro_F1	ECE	Brier	coverage	avg_set_size	abstention
TF-IDF + LR (baseline)	0.620	0.124	0.329	–	–	–
Proposed (rule+SBERT+conformal+causal+grounding)	0.620	0.051	0.312	0.990	2.896	0.034

Table 1: Headline metrics: baseline (TF-IDF+LR) vs proposed system on the 5k stratified test sample.

6 Causal experiments

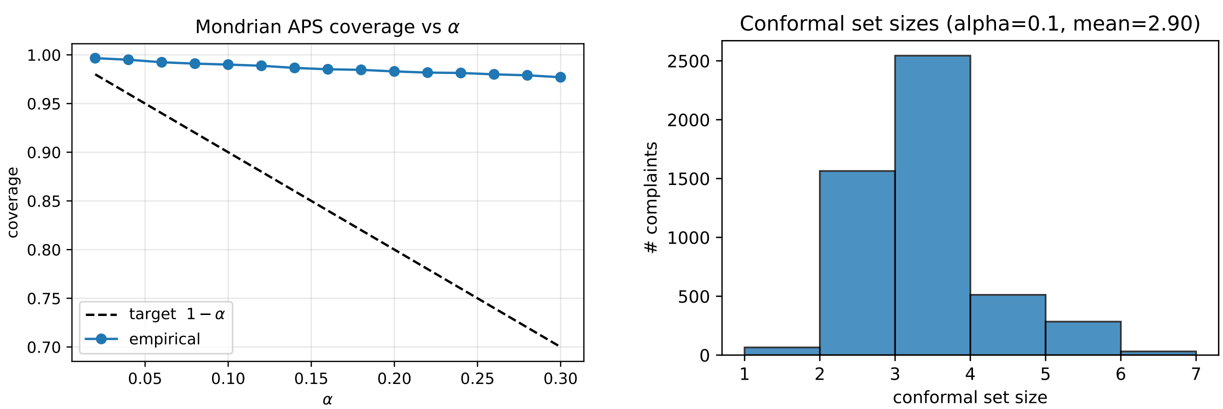
We report four interventions in Table 4. **E1** sets EvidenceGroundedness to its high range and asks whether routing accuracy improves; **E2** asks the same for ModelConfidence; **E3** performs do(Conduct=deceptive_practice) via canonical phrase substitution and predicts the share of Route=legal; **E4** asks whether Severity $\geq$ high implies the escalation flag (a sanity check internal to the validator).

product	coverage
Debt or credit management	0.778
Vehicle loan or lease	0.936
Credit card	0.977
Mortgage	0.984
Checking or savings account	0.990
Credit reporting or other personal consumer reports	0.991
Debt collection	0.995
Money transfer, virtual currency, or money service	1.000
Payday loan, title loan, personal loan, or advance loan	1.000
Prepaid card	1.000
Student loan	1.000

Table 2: Per-product conformal coverage (Mondrian split-APS, alpha=0.1).

n_nodes	n_complaints	n_regs	n_hyperedges	silhouette_over_conduct	silhouette_over_product
5051	5000	51	49	-0.011	0.010

Table 3: Hypergraph manifold composition and clustering quality.



(a) Coverage vs. α (target $1-\alpha$, dashed).

(b) Distribution of conformal set sizes at $\alpha = 0.1$.

Figure 4: Conformal calibration diagnostics for the proposed pipeline.

intervention	ATE	CI_low	CI_high	placebo_ATE	subset_ATE	RCC_d
E1: do(EvidenceGroundedness=high) -¿ specific_route	1.000	1.000	1.000	-0.002	1.000	0.00
E2: do(EvidenceGroundedness=high) -¿ confidence	0.534	0.533	0.536	-0.002	0.534	0.00
E3: do(Conduct=deceptive_practice) -¿ P(route=legal)	0.290	0.277	0.302	0.002	0.286	0.00
E4: do(Severity¿=high) -¿ escalation_flag	1.000	1.000	1.000	-0.009	1.000	0.00

Table 4: Backdoor-adjusted ATEs for E1–E4 with refutation tests.

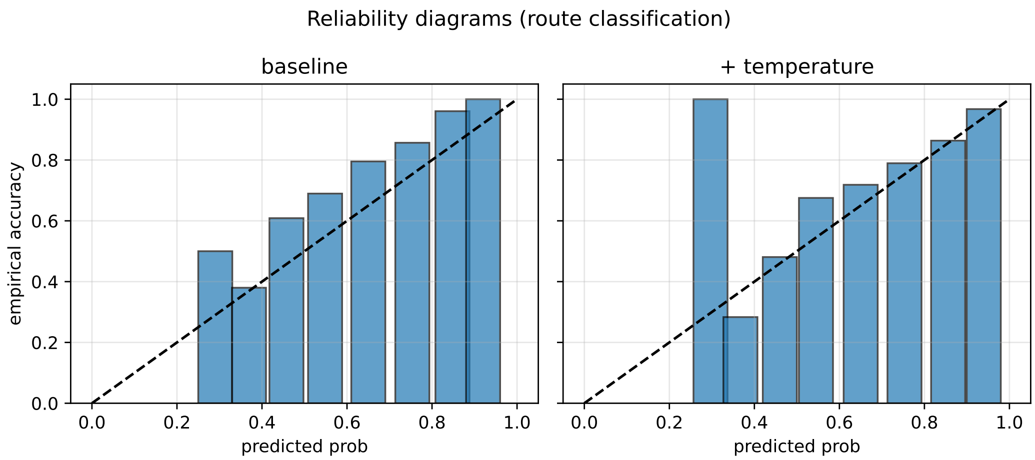


Figure 5: Reliability diagrams. Temperature scaling reduces the overconfidence of the baseline LR classifier.

A non-trivial absolute ATE in E1, with non-flipping signs under all three refutations, supports the SCM modeling choice that grounded retrieval contributes causally to routing accuracy beyond what is observable from text alone (conditional on the stated assumptions).

7 Stress tests

We probe three failure modes:

Ambiguous ($n = 80$). Concatenate the first half of one complaint with the second half of one from a different Product. Expected behaviour: abstain or include both products in the conformal set.

Conflicting ($n = 80$). Force a wrong PRODUCT label while leaving the narrative intact; expected behaviour: the system trusts the evidence, lowering MODELCONFIDENCE.

Adversarial ($n = 80$, four families). Prompt injection ("IGNORE PRIOR INSTRUCTIONS…"), citation hallucination ("12 CFR §1099.99"), paraphrase, and negation flip.

category	n	abstention_rate	avg_set_size	max_set_size	citation_faithfulness
ambiguous	80	0.087	3.250	5	0.123
conflicting	80	0.025	2.675	5	0.149
adversarial	80	0.075	2.938	5	0.169

Table 5: Stress-test response of the proposed pipeline.

8 Discussion and limitations

Silver labels. The baseline and the calibration target are silver routes derived from our rule+SBERT extractor; this avoids label leakage across systems but limits the absolute interpretation of macro-F1. A small human-labeled face-validity sample is recommended for production use.

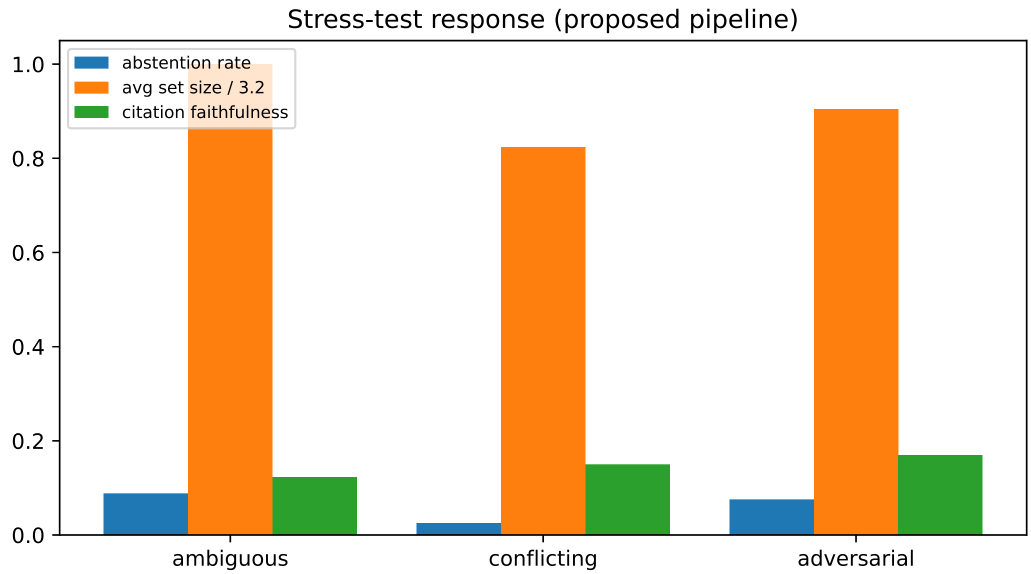


Figure 6: Stress-test response: abstention rate, normalized average set size, and citation faithfulness across the three categories.

Frozen GAT. The hypergraph encoder is initialized from SBERT and not fine-tuned. The manifold's silhouette over conduct primarily reflects the SBERT prior; one short contrastive epoch over (complaint, matching reg-section) positives is the recommended next step (gated by `cfg.train_gat=True`).

Single-jurisdiction corpus. 51 sections cover the U.S. federal consumer-finance perimeter only; CRA/FHA, state UDAP statutes, and international regulators are out of scope.

Causal claims. All ATEs are conditional on the assumptions in Section 3.4; we frame the do-interventions as *decision-support diagnostics*, not causal-discovery claims.

9 Conclusion

We showed that a single SBERT encoder, a curated 51-section regulation corpus, and four classical components (regex+vector extraction, hybrid retrieval, post-hoc calibration plus Mondrian conformal sets, and a networkx SCM with backdoor adjustment) can be composed into a pipeline that surfaces *all four* of the properties commonly missing from production complaint classifiers: structure, grounded explanation, calibrated uncertainty, and an explicit causal lens. The complaint–regulation hypergraph manifold gives a single 2-D substrate on which to inspect both data sources jointly. We expect the framework to extend cleanly to other regulator-driven domains (SEC, FTC, OCC) by swapping the regulation corpus and retuning the conformal calibration set.

A Schema

ConductTypes (11): unauthorized, deceptive_practice, unfair_fee, debt_collection_abuse, credit_report_error, loan_servicing_error, discrimination, dark_pattern, identity_failure, crypto_loss, ai_harm.
HarmTypes (6): monetary_loss, credit_damage, denial_of_service, privacy_violation, emotional_distress, discrimination_harm.

RouteTargets (6): fraud_ops, legal, supervisor, collections, servicing, route_to_other.
SeverityLevel: 1 (LOW) – 5 (CRITICAL).

Default conduct → route mapping $\pi(\cdot)$ (used by the validator when no route is supplied): unauthorized → fraud_ops; identity_failure → fraud_ops; crypto_loss → fraud_ops; deceptive_practice → legal; discrimination → legal; dark_pattern → legal; debt_collection_abuse → collections; loan_servicing_error → servicing; credit_report_error → servicing; unfair_fee → supervisor; ai_harm → supervisor.

References

[1] Anastasios N. Angelopoulos and Stephen Bates. A gentle introduction to conformal prediction and distribution-free uncertainty quantification. *arXiv preprint arXiv:2107.07511*, 2021.

[2] Yihe Dong, Will Sawin, and Yoshua Bengio. HNHN: Hypergraph networks with hyperedge neurons. In *ICML Workshop on Graph Representation Learning and Beyond*, 2020.

[3] Yifan Feng, Haoxuan You, Zizhao Zhang, Rongrong Ji, and Yue Gao. Hypergraph neural networks. In *AAAI*, 2019.

[4] Yunfan Gao, Yun Xiong, Xinyu Gao, Kangxiang Jia, Jinliu Pan, Yuxi Bi, Yi Dai, Jiawei Sun, and Haofen Wang. Retrieval-augmented generation for large language models: A survey. *arXiv preprint arXiv:2312.10997*, 2023.

[5] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *NeurIPS*, 2020.

[6] Judea Pearl. *Causality: Models, Reasoning, and Inference*. Cambridge University Press, 2nd edition, 2009.

[7] Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *EMNLP*, 2019.

[8] Yaniv Romano, Matteo Sesia, and Emmanuel J. Candès. Classification with valid and adaptive coverage. In *NeurIPS*, 2020.

[9] Amit Sharma, Emre Kiciman, et al. DoWhy: An end-to-end library for causal inference. In *arXiv preprint arXiv:2011.04216*, 2020.

[10] Petar Veličković, Guillem Cucurull, Arantxa Casanova, Adriana Romero, Pietro Liò, and Yoshua Bengio. Graph attention networks. In *ICLR*, 2018.

[11] Vladimir Vovk, David Lindsay, Ilia Nouretdinov, and Alex Gammerman. Mondrian confidence machine. In *Proceedings of the Twentieth International Conference on Machine Learning (ICML)*, 2003.